

T'Shara Haynes

AI/ML Research / Data Analytics / Sustainable Development

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PROFESSIONAL SUMMARY

Data Analyst and ML researcher with a strong foundation in Computer Science and Environmental Science. Passionate about applying data-driven approaches to environmental challenges, including hazard forecasting and sustainable development. Proficient in machine learning, spatial analysis and data visualization. Hands-one experience building predictive models from complex environmental and financial datasets. Committed to using data to inform equitable policy decisions and drive social impact.

EDUCATION

University of the West Indies, Cave Hill Campus — BSc. Computer Science with Environmental Science
Relevant Coursework: Data Structures & Algorithms, Environmental Ethics, Environmental Health, Earth's Climate

WORK EXPERIENCE

TRINRE Insurance – Port-of-Spain, Trinidad and Tobago June 2026- August 2026
Information Communications Technology Intern

- Mapped object-relational data models for the Finance Module, streamlining database queries and improving system efficiency.
- Participated as a key organiser in the compatibility testing of the company’s TSQL versions for the purpose of facilitating the update of TRINRE’S TSQL version from 2014 to 2017.
- Audited and corrected Account Schedules in Dynamics 365, ensuring Trial Balance alignment and financial reporting accuracy.

Tessera Studios – Remote, Barbados February 2026 – Present
Contract Project Assistant

- Conducted background research on industry trends and relevant technologies to inform project directions.
- Wrote engaging content for social media and blogs while maintaining brand voice and clarity.
- Evaluated project concepts through user interviews and feedback analysis.
- Built user-facing components using agreed technologies and frameworks for clients.

Central Bank of Barbados - Bridgetown, Barbados June 2025- August 2025
Governor’s Office Intern

- Mapped end-to-end payment system workflows using system documentation to support integration for the BiMPay rollout.
- Produced clear process documentation to enable cross-team implementation.
- Reviewed regional data protection standards to ensure applicability of compliant use of AI and customer data.
- Authored a research paper on fraud detection in instant payment systems contributing insights on machine learning applications in low-label financial environments.

Pellucid IT Inc - Remote, Barbados October 2024- May 2025
Project Assistant

- Documented team workflows and improved task tracking to keep projects organized and on schedule.
- Implemented Scrum Methodology to enhance project agility and streamline workflows.
- Boosted social media engagement by 20% through a targeted content strategy and community interaction.
- Tracked timelines, identified blockers, and kept stakeholders informed to ensure smooth project delivery.
- Coordinated scheduling between team members and external stakeholders.

PUBLICATIONS & PRESENTATIONS

- Towards the Preservation of Caribbean Sign Languages Using AI" *Accepted - ICCHP 2026 | Invited for publication via Springer*
- Fraud Detection in Instant Payment Systems: A Caribbean Case Study of Machine Learning Constraints and Opportunities. *Presented - Central Bank of Barbados Annual Review Seminar, 2025*
- When Data is Sparse: Machine Learning Approaches to Air Pollutant Forecasting in Trinidad and Tobago. *Accepted – Caribbean STEM Engine for Economic Development Summit, 2026*

TECHNICAL SKILLS

- **Languages:** Python, SQL, C++, JavaScript, HTML/CSS, PHP, MATLAB
- **Data & ML:** Pandas, Scikit-Learn, Tensorflow, PyTorch, Power BI, Matplotlib
- **GI & Spatial:** QGIS
- **Tools & Platforms:** Git, Microsoft Office Suite, Asana, Trello
- **Databases:** MySQL, MongoDB
- **Design Tools:** Canva, Figma, Adobe Lightroom, Gimp

ADDITIONAL CERTIFICATES

- AWS

 - Getting Started with Security
 - Introduction to Cloud 101
- QGIS

 - Fundamentals of GIS

TECHNICAL PROJECTS

- Sign Language Recognition System

Python | TensorFlow | OpenCV

Research Project

 - Trained a machine learning model using Google’s Teachable Machine to classify sign language gestures.
 - Pre-processed and labelled dataset; performed feature extraction to produce a ~90% accuracy.
 - Implemented real-time inference pipeline using OpenCV and Python
 - Designed to reduce communication barriers between deaf and hearing individuals.
- Fraud Detector for Instant Payment Systems

Python | Streamlit | SciKit-Learn

Research Project

 - Built a transaction simulation app to model financial payment flows.
 - Applied SciKit-Learn feature engineering and model evaluation techniques to detect anomalies in transaction data (~92% accuracy).
 - Focused on challenges in low-labelled Caribbean IPS environments.
- Data Augmentation for Air Pollution Forecasting

Python | Matplotlib | SciKit-Learn

Research Project

 - Built a 30-day air pollutant forecasting system for Trinidad and Tobago monitoring stations.
 - Trained and evaluated KNN, XGBoost and Support Vector Regression models using time-series cross-validation to prevent data leakage.